

ARTIFICIAL INTELLIGENCE



CHAPTER-7

Genetic Algorithm



Introduction of Genetic Algorithm

- Genetic Algorithm (GA) is a search-based optimization technique based on the principles of **Genetics and Natural Selection**. It is frequently used to find optimal or near-optimal solutions to difficult problems which otherwise would take a lifetime to solve. It is frequently used to solve optimization problems, in research, and in machine learning.

Optimization is the process of **making something better**. In any process, we have a set of inputs and a set of outputs as shown in the following figure.



Introduction of Genetic Algorithm

Optimization refers to finding the values of inputs in such a way that we get the “best” output values. The definition of “best” varies from problem to problem, but in mathematical terms, it refers to maximizing or minimizing one or more objective functions, by varying the input parameters.

The set of all possible solutions or values which the inputs can take make up the search space. In this search space, lies a point or a set of points which gives the optimal solution. The aim of optimization is to find that point or set of points in the search space.





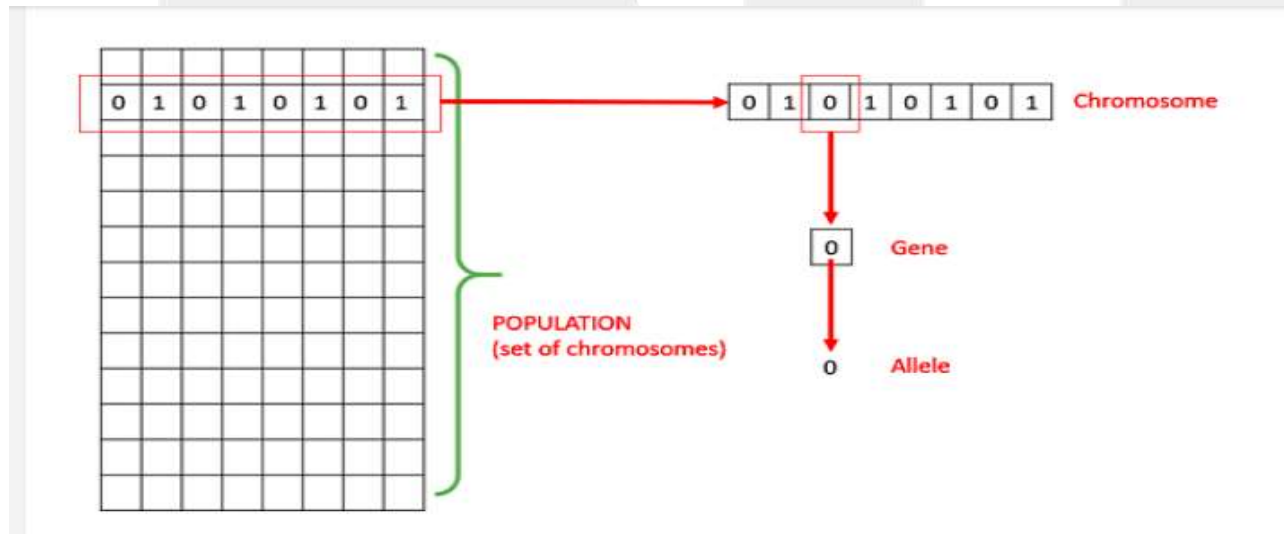
Basic Terminology

Population – It is a subset of all the possible (encoded) solutions to the given problem.

Chromosomes – A chromosome is one such solution to the given problem.

Gene – A gene is one element position of a chromosome.

Allele – It is the value a gene takes for a particular chromosome.





Basic Terminology

Genotype – Genotype is the population in the computation space. In the computation space, the solutions are represented in a way which can be easily understood and manipulated using a computing system

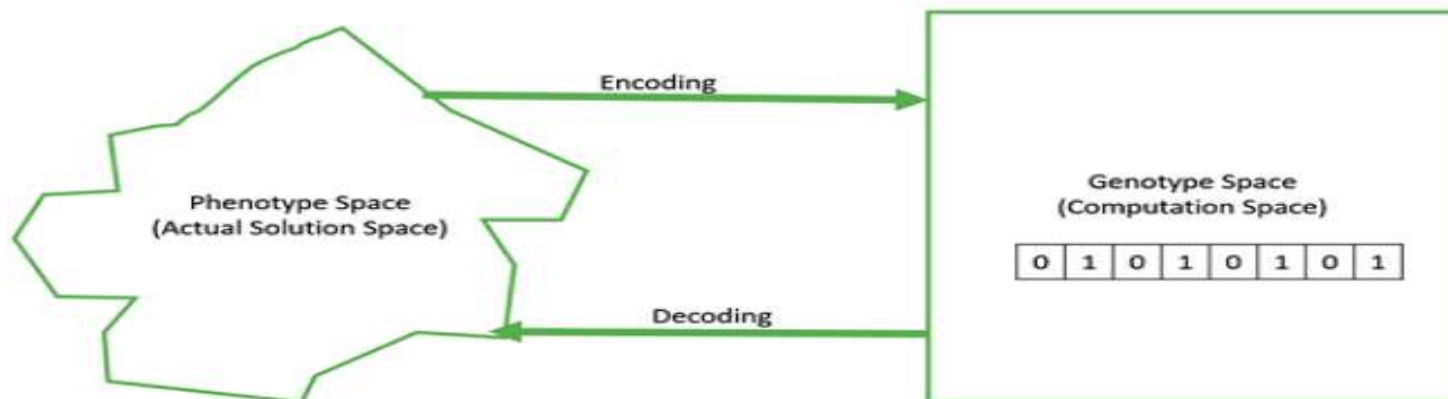
Phenotype – Phenotype is the population in the actual real world solution space in which solutions are represented in a way they are represented in real world situations.

Decoding and Encoding - Decoding is a process of transforming a solution from the genotype to the phenotype space, while encoding is a process of transforming from the phenotype to genotype space.





Basic Terminology



- **Fitness Function** – A fitness function simply defined is a function which takes the solution as input and produces the suitability of the solution as the output. In some cases, the fitness function and the objective function may be the same, while in others it might be different based on the problem.





ADVANTAGES of GA

- Does not require any derivative information (which may not be available for many real-world problems).
- Is faster and more efficient as compared to the traditional methods.
- Has very good parallel capabilities.
- Optimizes both continuous and discrete functions and also multi-objective problems.
- Provides a list of “good” solutions and not just a single solution.
- Always gets an answer to the problem, which gets better over the time.
- Useful when the search space is very large and there are a large number of parameters involved.



DISADVANTAGES of GA

GAs are not suited for all problems, especially problems which are simple and for which derivative information is available.

Fitness value is calculated repeatedly which might be computationally expensive for some problems.

Being stochastic, there are no guarantees on the optimality or the quality of the solution.

If not implemented properly, the GA may not converge to the optimal solution.

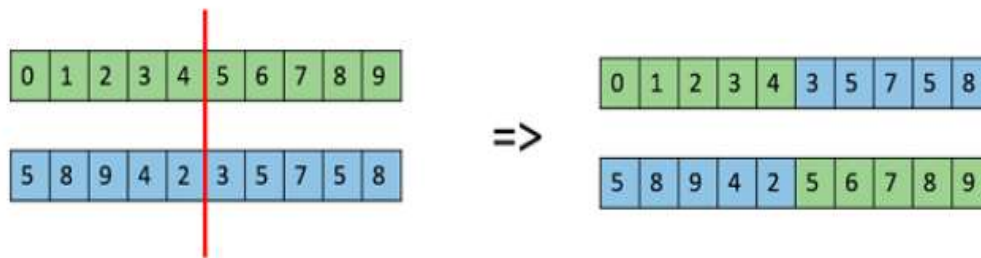


Operator of Genetic algorithm

- **Crossover Operators :-**there are different type of Crossover Operators

One Point Crossover

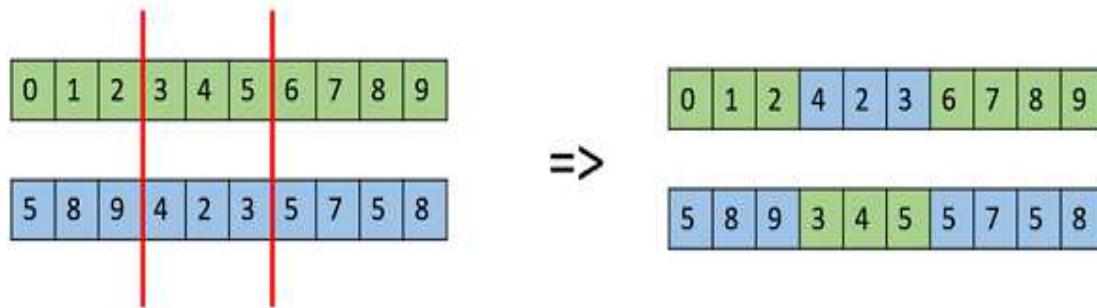
In this one-point crossover, a random crossover point is selected and the tails of its two parents are swapped to get new off-springs.



Operator of Genetic algorithm

Multi Point Crossover

Multi point crossover is a generalization of the one-point crossover where in alternating segments are swapped to get new off-springs.



Operator of Genetic algorithm

Uniform Crossover

In a uniform crossover, we don't divide the chromosome into segments, rather we treat each gene separately. In this, we essentially flip a coin for each chromosome to decide whether or not it'll be included in the off-spring. We can also bias the coin to one parent, to have more genetic material in the child from that parent.

0	1	2	3	4	5	6	7	8	9
---	---	---	---	---	---	---	---	---	---

5	8	9	4	2	3	5	7	5	8
---	---	---	---	---	---	---	---	---	---

=>

5	1	9	4	4	5	5	7	5	9
---	---	---	---	---	---	---	---	---	---

0	8	2	3	2	3	6	7	8	8
---	---	---	---	---	---	---	---	---	---

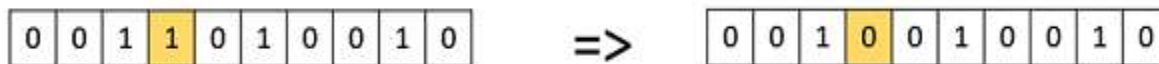


Operator of Genetic algorithm

Mutation Operators

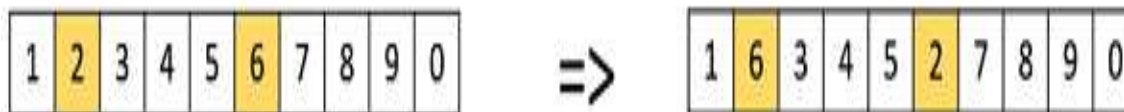
Bit Flip Mutation

In this bit flip mutation, we select one or more random bits and flip them. This is used for binary encoded GAs.



Swap Mutation

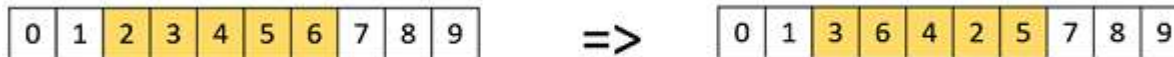
In swap mutation, we select two positions on the chromosome at random, and interchange the values. This is common in permutation based encodings.



Operator of Genetic algorithm

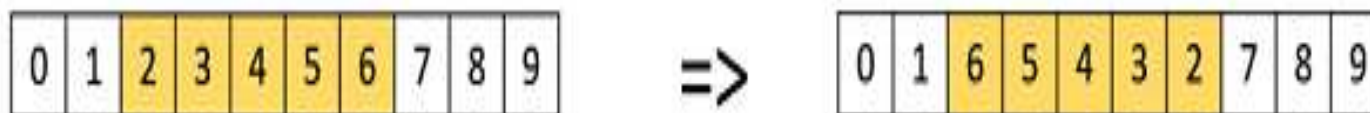
Scramble Mutation

.Scramble mutation is also popular with permutation representations. In this, from the entire chromosome, a subset of genes is chosen and their values are scrambled or shuffled randomly.



Inversion Mutation

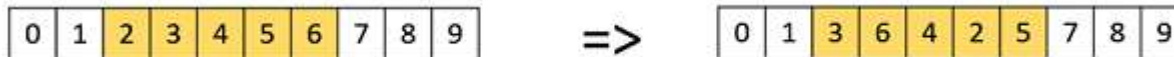
In inversion mutation, we select a subset of genes like in scramble mutation, but instead of shuffling the subset, we merely invert the entire string in the subset.



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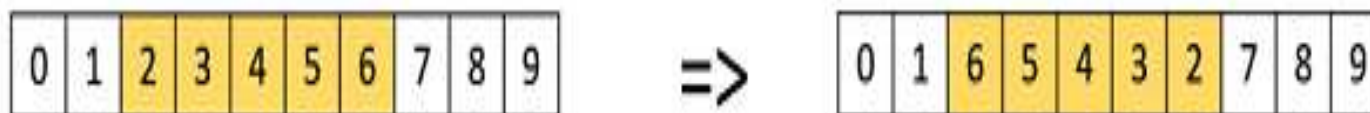
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Inversion Mutation

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Operator of Genetic algorithm

3. Parent Selection Operator

Parent Selection is the process of selecting parents which mate and recombine to create off-springs for the next generation. Parent selection is very crucial to the convergence rate of the GA as good parents drive individuals to a better and fitter solutions.

Fitness Proportionate Selection

Fitness Proportionate Selection is one of the most popular ways of parent selection. In this every individual can become a parent with a probability which is proportional to its fitness. Therefore, fitter individuals have a higher chance of mating and propagating their features to the next generation. Therefore, such a selection strategy applies a selection pressure to the more fit individuals in the population, evolving better individuals over time.

There are different selection method



Operator of Genetic algorithm

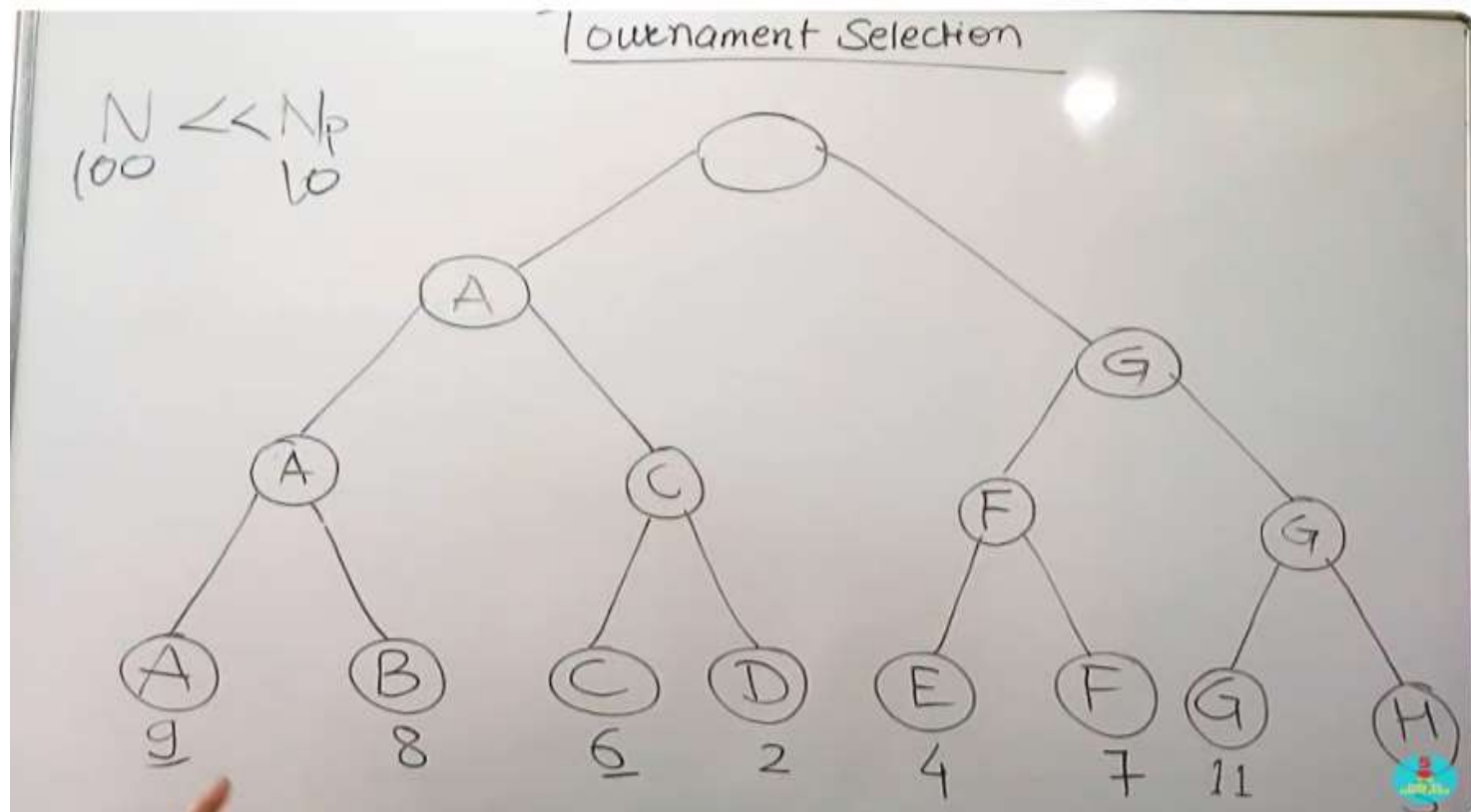
Tournament Selection

Tournament Selection is a Selection Strategy used for selecting the fittest candidates from the current generation in a Genetic Algorithm. These selected candidates are then passed on to the next generation. In a K-way tournament selection, we select k-individuals and run a tournament among them. Only the fittest candidate amongst those selected candidates is chosen and is passed on to the next generation. In this way many such tournaments take place and we have our final selection of candidates who move on to the next generation.

If the tournament size is larger, weak candidates have a smaller chance of getting selected as it has to compete with a stronger candidate



Operator of Genetic algorithm





Operator of Genetic algorithm

Rank Based selection

In a genetic algorithm, rank selection is a selection method that ranks the population of individuals based on their fitness and then assigns fitness values to each chromosome based on their rank:

Rank the population: Sort the population of individuals by fitness

Assign fitness values: Assign fitness values to each chromosome based on their rank, with the worst chromosome receiving a fitness of 1, the second worst a fitness of 2, and so on

Select chromosomes: Select chromosomes based on their fitness values

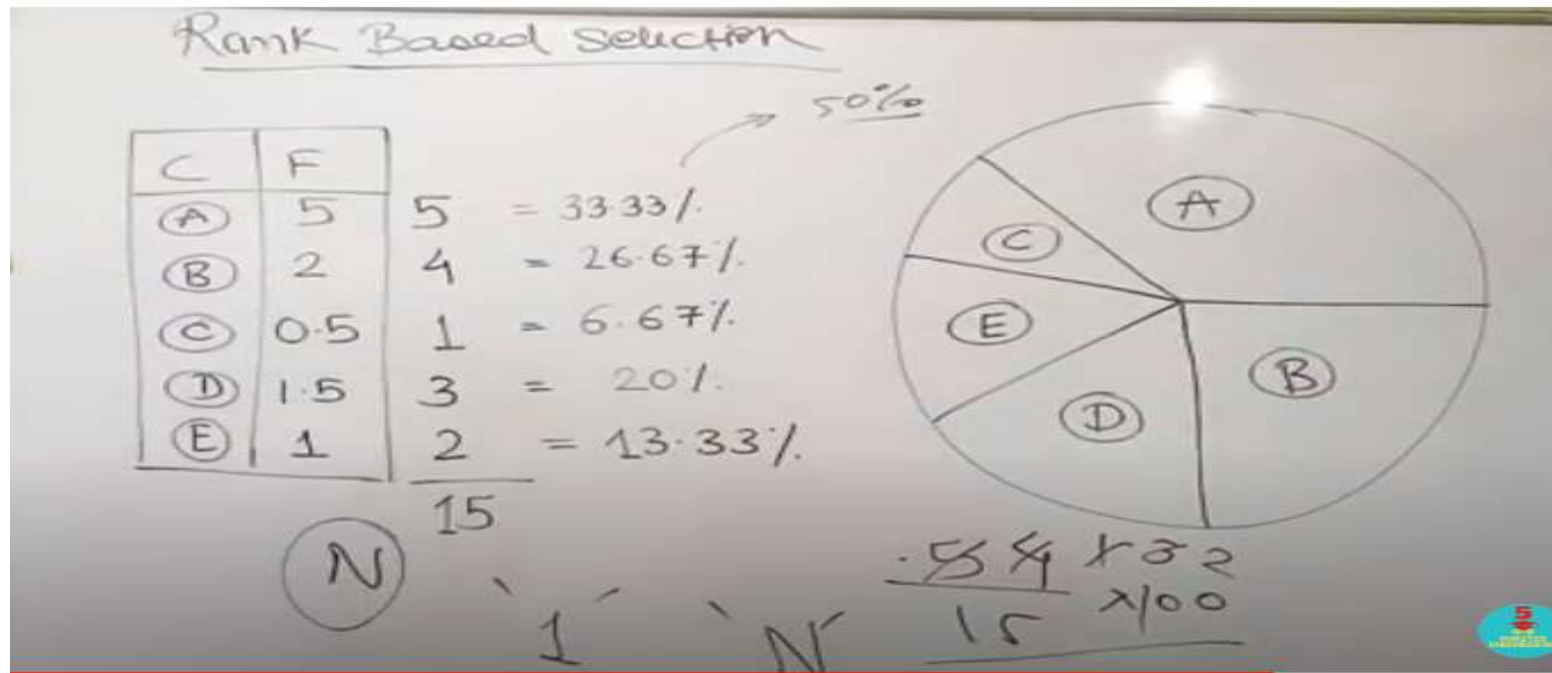
Rank selection can help ensure that all chromosomes have a chance to be selected, even if the fitnesses of the chromosomes differ significantly. However, it can also lead to slower convergence if the best chromosomes are not much different from the others.





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Rank Based selection

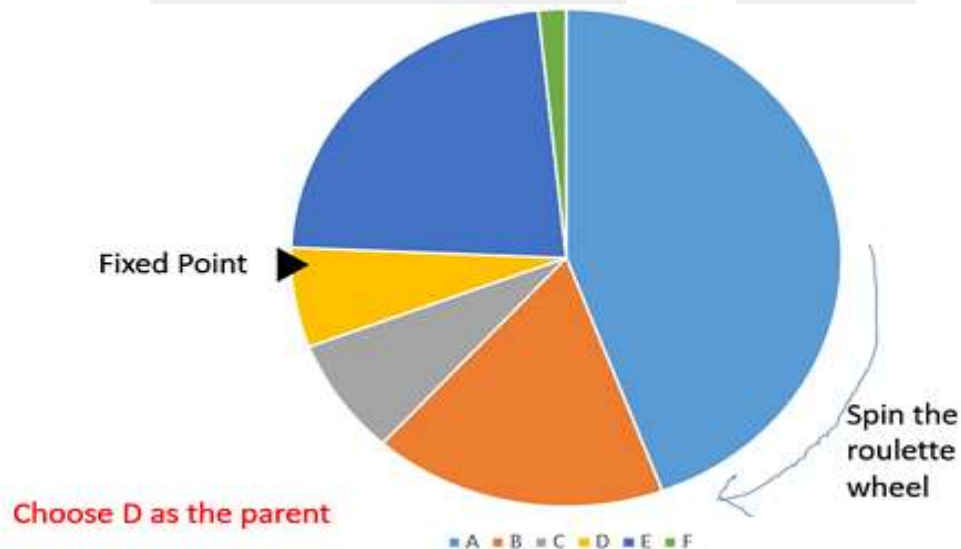




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Roulette Wheel Selection

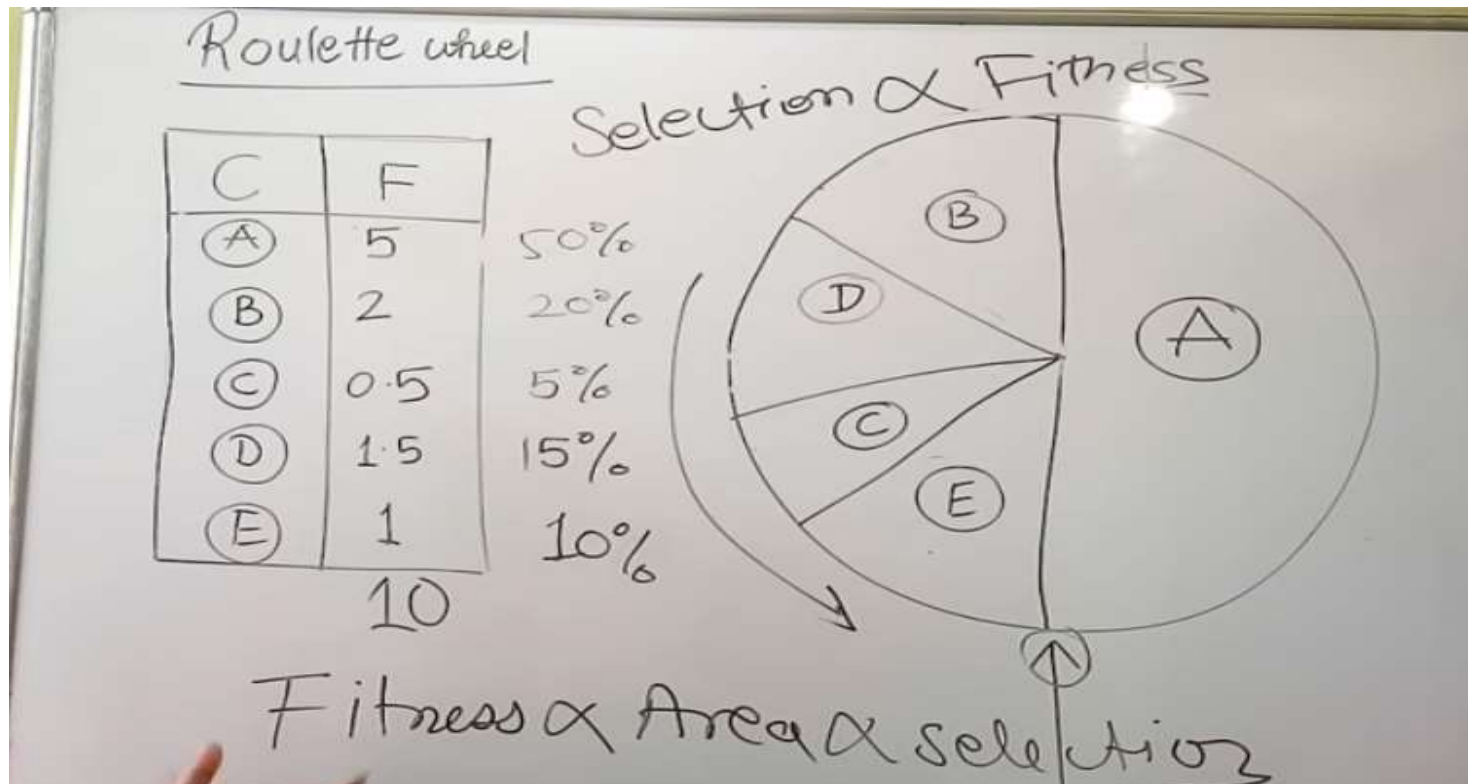
In a roulette wheel selection, the circular wheel is divided as described before. A fixed point is chosen on the wheel circumference as shown and the wheel is rotated. The region of the wheel which comes in front of the fixed point is chosen as the parent. For the second parent, the same process is repeated.



Chromosome	Fitness Value
A	8.2
B	3.2
C	1.4
D	1.2
E	4.2
F	0.3



Operator of Genetic algorithm





Cycle of Genetic algorithm

The **genetic algorithm cycle** in Artificial Intelligence (AI) is a search heuristic that mimics the process of natural selection to solve optimization and search problems. The cycle typically involves the following key steps:

1. Initialization

A population of candidate solutions, called chromosomes, is randomly generated. These solutions represent possible answers to the problem at hand.

2. Selection

The fitness of each chromosome is evaluated using a fitness function that measures how well each solution solves the problem.

Solutions with higher fitness have a higher chance of being selected for reproduction.





Cycle of Genetic algorithm

3. Crossover (Recombination)

Selected chromosomes (parents) are paired, and a crossover operation is applied. During this operation, segments of the parents' solutions are combined to create new offspring, introducing variation into the population.

4. Mutation

A small, random alteration is made to some offspring to introduce new genetic material. This helps prevent the algorithm from getting stuck in local optima and encourages exploration of the solution space.

5. Evaluation

The fitness of the new offspring is evaluated using the same fitness function.

6. Replacement

The new generation of solutions replaces the old generation. In some versions of genetic algorithms, only the best solutions (elite solutions) from the old generation may be carried over to the new generation.



Cycle of Genetic algorithm

7. Termination

The process repeats for a specified number of generations or until a stopping condition (e.g., a solution with sufficient fitness or convergence) is met.



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Function Optimization of GA

Function optimization using a **Genetic Algorithm (GA)** in AI refers to finding the optimal solution (maximum or minimum) for a given function by simulating the process of natural evolution. Genetic algorithms are heuristic search algorithms based on the principles of natural selection and genetics. Here's a breakdown of how function optimization works using GA.

Steps of Genetic Algorithm for Function Optimization

1. Representation of the Problem
2. Initialization of Population:
3. Fitness Function:
4. Selection:
5. Crossover (Recombination):
6. Mutation:
7. Replacement:
8. Termination:



Edge recombination operator (ERO)

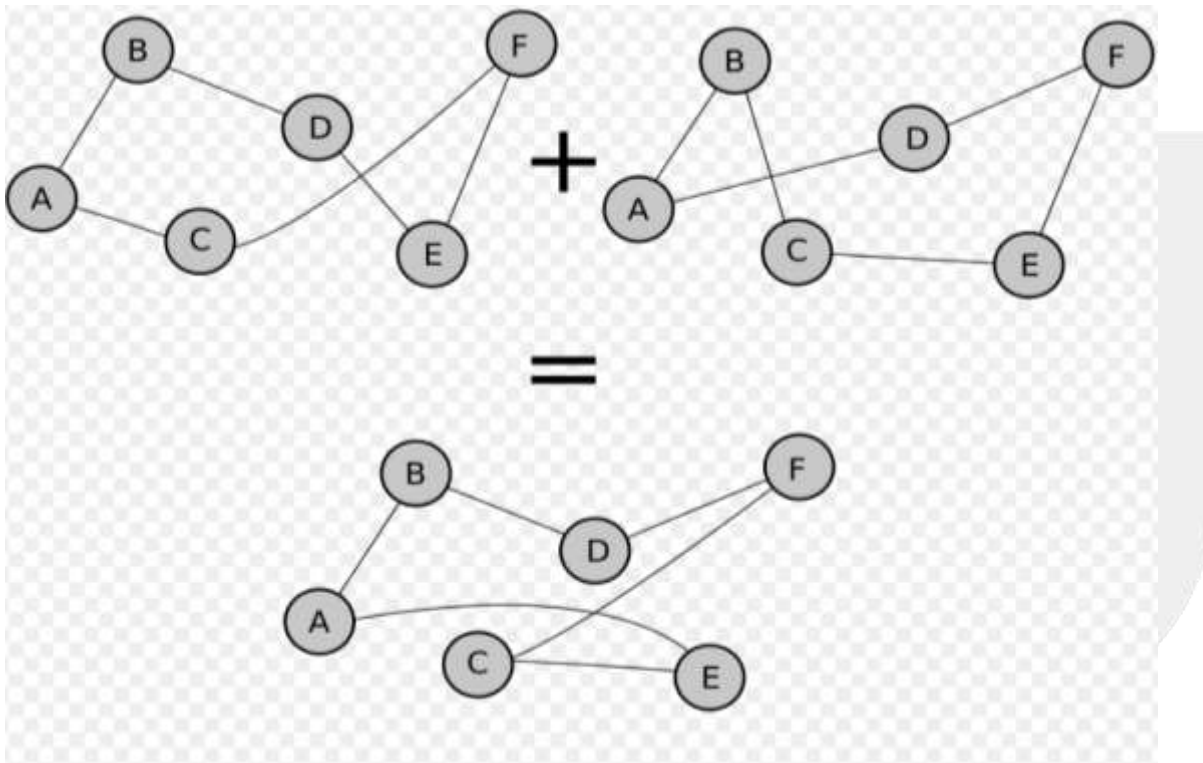
The edge recombination operator (ERO) is an operator that creates a path that is similar to a set of existing paths (parents) by looking at the edges rather than the vertices. The main application of this is for crossover in genetic algorithms when a genotype with non-repeating gene sequences is needed such as for the travelling salesman problem. It was described by Darrell Whitley and others in 1989.

ERO is based on an adjacency matrix, which lists the neighbors of each node in any parent.

For example, in a travelling salesman problem such as the one depicted, the node map for the parents CABDEF and ABCEFD (see illustration) is generated by taking the first parent, say, 'ABCEFD' and recording its immediate neighbors, including those that roll around the end of the string



Function Optimization of GA



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